

ZHIYUAN ZHA

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👤 WORK EXPERIENCE

- Jilin University, China** 04/2024 -- Now
Elite Full Professor (tenured)
- Nanyang Technological University, Singapore** 07/2023 -- 04/2024
Senior Research Fellow at the Rapid-Rich Object Search Lab (ROSE Lab), working with Prof. Alex Chichung Kot (IEEE Life Fellow) and Prof. Bihan Wen
- Nanyang Technological University, Singapore** 08/2019 -- 06/2023
Research Fellow at the Rapid-Rich Object Search Lab (ROSE Lab), working with Prof. Alex Chichung Kot (IEEE Life Fellow) and Prof. Bihan Wen
- University of Electronic Science and Technology of China, China** 08/2018 -- 08/2020
Postdoctoral Working with Prof. Ce Zhu (IEEE/OSA Fellow)

🎓 EDUCATION

- Nanjing University, Nanjing, China** 08/2015 -- 06/2018
Ph.D. in Electrical and Electronic Engineering (EEE)
- Kunming University of Science and Technology, Kunming, China** 09/2012 -- 06/2015
M.S. in Computer Science (CS)
- Jiangsu University, Zhenjiang, China** 09/2006 -- 11/2010
B.S. in Computer Science (CS)

🔍 RESEARCH INTERESTS

Computer Vision, Image Super-Restoration, Artificial Intelligence.

🎁 HONORS AND AWARDS

- **IEEE Senior Member**
- **CCF Senior Member**
- **CSIG Senior Member**
- *The 20th IEEE International Conference on Multimedia and Expo* 07/2020
Best Paper Runner Up Award **Top 0.5%**
- *The 17th IEEE International Conference on Multimedia and Expo* 07/2017
Platinum Best Paper Award **Top 0.5%**

🎵 ACADEMIC SERVICE

- **Associate Editor** of **IEEE Transactions on Circuits and Systems for Video Technology**, 01/2026-12/2027
- **Young Affiliate of Associate Editor** of **Machine Intelligence Research**, 01/2026-01/2027
- **Associate Editor** of **IEEE Transactions on Image Processing**, 07/2024-Present
- **Lead Guest Editor** of **IEEE Journal of Selected Topic in Signal Processing**, 01/2025-Present
- **Handing Editor** of **Signal Processing**, 07/2024-Present
- **Associate Editor** of **The Visual Computer**, 04/2023-Present
- **IEEE Computational Imaging Technical Committee (CI TC) Member**, 01/2024-01/2027

- Area Chair of IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2025-2026)
- Area Chair of the 5th IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR 2022)
- Session Chair of the 47th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2022)

§ FUNDING

- 1) National Natural Science Foundation of China, Excellent Young Scientists Fund Program, 01/2025-12/2028, **Principal Investigator**
- 2) National Natural Science Foundation of China, General Program, 01/2025-12/2028, **Principal Investigator**
- 3) Ministry of Human Resources and Social Security, National Foreign Experts Project, 01/2025-12/2026, **Principal Investigator**

♡ JOURNAL

1. Hengmin Zhang, Jian Yang, Jianjun Qian, Guangwei Gao, Xiangyuan Lan, **Zhiyuan Zha**, Bihan Wen, "Efficient Image Classification via Structured Low-Rank Matrix Factorization Regression," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 1496-1509, 2024. (IF: 8.0)
2. Kunlei Jing, Hebo Ma, Chen Zhang, **Zhiyuan Zha**, and Bihan Wen, A regularized deep self-expression feature augmentation network for few-shot unconstrained palmprint recognition, *Pattern Recognition*, vol. 173, pp. 112904, 2026. (IF: 7.60)
3. Haonan Zhang, Longjun Liu, Fei Hui, Bo Zhang, Hengmin Zhang and **Zhiyuan Zha**. "CLEAN: Category Knowledge-Driven Compression Framework for Efficient 3D Object Detection", *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, vol. 47, no. 10, pp. 8740-8755, 2025. (IF: 18.60)
4. Wudi Chen, **Zhiyuan Zha***, Shigang Wang, Liaqat Ali, Bihan Wen, Xin Yuan, Jiantao Zhou and Ce Zhu. "Texture-Consistent 3D Scene Style Transfer via Transformer-Guided Neural Radiance Fields", *IEEE Transactions on Image Processing (TIP)*, vol. 34, pp. 7193-7208, 2025. (IF: 13.70, *Corresponding Author)
5. Sirui Pan, **Zhiyuan Zha***, Shigang Wang, Yue Li, Zipei Fan, Gang Yan, Binh T. Nguyen, Bihan Wen and Ce Zhu. "Triply Laplacian Scale Mixture Modeling for Seismic Data Noise Suppression", *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, vol. 63, pp. 1-14, 2025. (IF: 8.6, *Corresponding Author)
6. Sirui Pan, **Zhiyuan Zha***, Shigang Wang, Bihan Wen, Zipei Fan, Gang Yan, Binh T. Nguyen and Yue Li. "Joint Tensor Ring Decomposition and Unidirectional Total Variation for Seismic Data Denoising", *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, vol. 63, pp. 1-15, 2025. (IF: 8.6, *Corresponding Author)
7. **Zhiyuan Zha**, Bihan Wen, Xin Yuan, Jiachao Zhang, Jiantao Zhou, Yilong Lu and Ce Zhu. "Nonlocal Structured Sparsity Regularization Modeling for Hyperspectral Image Denoising", *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, vol. 61, pp. 1-16, 2023. (IF: 8.125)
8. **Zhiyuan Zha**, Bihan Wen, Xin Yuan, Jiachao Zhang, Jiantao Zhou and Ce Zhu. "Structured residual sparsity for video compressive sensing reconstruction", *Signal Processing*, 109513, 2024. (IF: 4.4)
9. **Zhiyuan Zha**, Bihan Wen, Xin Yuan, Saiprasad Ravishankar, Jiantao Zhou and Ce Zhu. "Learning Non-local Sparse and Low-Rank Models for Image Compressive Sensing", *IEEE Signal Processing Magazine (SPM)*, vol. 40, no. 1, pp. 32-44, Jan. 2023. (IF: 15.204)
10. **Zhiyuan Zha**, Bihan Wen, Xin Yuan, Joey Tianyi Zhou, Jiantao Zhou and Ce Zhu. "Triply Complementary Priors for Image Restoration", *IEEE Transactions on Image Processing (TIP)*, vol. 30, pp. 5819-5834, 2021. (IF: 10.856)
11. **Zhiyuan Zha**, Bihan Wen, Xin Yuan, Jiantao Zhou and Ce Zhu. "Image Restoration via Reconciliation of Group Sparsity and Low-Rank Models", *IEEE Transactions on Image Processing (TIP)*, vol. 30, pp. 5223-5238, 2021. (IF: 10.856)
12. **Zhiyuan Zha**, Xin Yuan, Bihan Wen, Jiantao Zhou and Ce Zhu. "Group Sparsity Residual Constraint With Non-Local Priors for Image Restoration", *IEEE Transactions on Image Processing (TIP)*, vol. 29, pp. 8960-8975, 2020. (IF: 9.340)
13. **Zhiyuan Zha**, Xin Yuan, Jiantao Zhou, Ce Zhu and Bihan Wen. "Image restoration via simultaneous nonlocal self-similarity priors", *IEEE Transactions on Image Processing (TIP)*, vol. 29, pp. 8561-8576,

2020. (IF: 9.340)

14. **Zhiyuan Zha**, Xin Yuan, Bihan Wen, Jiachao Zhang, Jiantao Zhou and Ce Zhu. "Image restoration using joint patch-group-based sparse representation", *IEEE Transactions on Image Processing (TIP)*, vol. 29, pp. 7735-7750, 2020. (IF: 9.340)
15. **Zhiyuan Zha**, Xin Yuan, Bihan Wen, Jiantao Zhou, Jiachao Zhang and Ce Zhu. "A benchmark for sparse coding: When group sparsity meets rank minimization", *IEEE Transactions on Image Processing (TIP)*, vol. 29, pp. 5094-5109, 2020. (IF: 9.340)
16. **Zhiyuan Zha**, Xin Yuan, Bihan Wen, Jiantao Zhou, Jiachao Zhang and Ce Zhu. "From rank estimation to rank approximation: Rank residual constraint for image restoration", *IEEE Transactions on Image Processing (TIP)*, vol. 29, pp. 3254-3269, 2020. (IF: 9.340)
17. **Zhiyuan Zha**, Bihan Wen, Xin Yuan, Jiantao Zhou, Ce Zhu and Alex Chichung Kot. "Low-Rankness Guided Group Sparse Representation for Image Restoration", *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2022. (IF: 14.255)
18. **Zhiyuan Zha**, Bihan Wen, Xin Yuan, Jiantao Zhou, Ce Zhu and Alex Chichung Kot. "A Hybrid Structural Sparsification Error Model for Image Restoration", *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, vol. 33, no. 9, pp. 4451-4465, 2022. (IF: 14.255)
19. **Zhiyuan Zha**, Bihan Wen, Xin Yuan, Jiachao Zhang, Jiantao Zhou, Xudong Jiang and Ce Zhu. "Multiple Complementary Priors for Multispectral Image Compressive Sensing Reconstruction", *IEEE Transactions on Cybernetics (TCYB)*, 2023. (IF: 19.118)
20. **Zhiyuan Zha**, Xin Yuan, Bihan Wen, Jiachao Zhang and Ce Zhu. "Non-Convex Structural Sparsity Residual Constraint for Image Restoration", *IEEE Transactions on Cybernetics (TCYB)*, vol. 52, no. 11, pp. 12440-12453, Nov. 2022. (IF: 19.118)
21. **Zhiyuan Zha**, Xinggan Zhang, Yu Wu, Qiong Wang, Xin Liu, Lan Tang and Xin Yuan. "Non-convex weighted Lp nuclear norm based ADMM framework for image restoration", *Neurocomputing*, vol. 311, pp. 209-224, 2018. (IF: 3.241)
22. **Zhiyuan Zha**, Xinggan Zhang, Qiong Wang, Lan Tang and Xin Liu. "Group-based sparse representation for image compressive sensing reconstruction with non-convex regularization", *Neurocomputing*, vol. 296, pp. 55-63, 2018. (IF: 3.241)
23. **Zhiyuan Zha**, Xinggan Zhang, Qiong Wang, Yechao Bai, Yang Chen, Lan Tang and Xin Liu. "Group sparsity residual constraint for image denoising with external nonlocal self-similarity prior", *Neurocomputing*, vol. 275, pp. 2294-2306, 2018. (IF: 3.241)
24. **Zhiyuan Zha**, Xin Liu, Xinggan Zhang, Yang Chen, Lan Tang, Yechao Bai, Qiong Wang and Zhenhong Shang. "Compressed sensing image reconstruction via adaptive sparse nonlocal regularization", *The Visual Computer*, vol. 34, pp. 117-137, 2018. (IF: 1.036)
25. Qiong Wang, Xinggan Zhang, Yu Wu, Lan Tang and **Zhiyuan Zha***. "Nonconvex Weighted Lp Minimization Based Group Sparse Representation Framework for Image Denoising", *IEEE Signal Processing Letters (SPL)*, vol. 24, pp. 1686-1690, Nov. 2017. (IF: 2.528)
26. Lanqing Guo, **Zhiyuan Zha**, Saiprasad Ravishankar, and Bihan Wen. "Exploiting Non-Local Priors via Self-Convolution for Highly-Efficient Image Restoration", *IEEE Transactions on Image Processing (TIP)*, vol. 31, pp. 1311-1324, 2022. (IF: 11.041)

♥ CONFERENCE

1. Wudi Chen, **Zhiyuan Zha***, Shigang Wang, Bihan Wen, Xin Yuan, Jiantao Zhou, zipei fan, Gang Yan, Ce Zhu. "KaRF: Weakly-Supervised Kolmogorov-Arnold Networks-based Radiance Fields for Local Color Editing", in *39th Annual Conference on Neural Information Processing Systems (NeurIPS)*, San Diego, USA, 2025. (*Corresponding Author, Poster)
2. **Zhiyuan Zha**, Bihan Wen, Xin Yuan, Jiantao Zhou and Ce Zhu. "Hyperspectral Image Denoising via Nonlocal Rank Residual Modeling", in *48th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Greece, pp. 1-5, 2023. (Poster)
3. **Zhiyuan Zha**, Bihan Wen, Xin Yuan, Jiantao Zhou and Ce Zhu. "Simultaneous Nonlocal Low-Rank and Deep Priors for Poisson Denoising", in *47th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Singapore, pp. 2320-2324, 2022. (Oral)
4. **Zhiyuan Zha**, Bihan Wen, Xin Yuan, Jiantao Zhou and Ce Zhu. "Low-Rank Regularized Joint Sparsity for Image Denoising", in *28th IEEE International Conference on Image Processing (ICIP)*, Anchorage, Alaska, USA, pp. 1644-1648, 2021. (Oral)

5. **Zhiyuan Zha**, Bihan Wen, Xin Yuan, Jiantao Zhou and Ce Zhu. "Reconciliation of Group Sparsity and Low-Rank Models for Image Restoration", in *20th IEEE International Conference on Multimedia and Expo(ICME)*, London, United Kingdom, pp. 1-6, 2020. (**Best Paper Runner Up Award, Top 0.5%, Total 834**, Oral)
6. **Zhiyuan Zha**, Xin Yuan, Jiantao Zhou, Ce Zhu and Bihan Wen. "A Hybrid Structural Sparse Error Model for Image Deblocking", in *45th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Barcelona, Spain, pp. 2493-2497, 2020. (Oral)
7. **Zhiyuan Zha**, Xin Yuan, Jiantao Zhou, Joey Tianyi Zhou, Bihan Wen and Ce Zhu. "The Power Of Triply Complementary Priors For Image Compressive Sensing", in *27th IEEE International Conference on Image Processing (ICIP)*, Abu Dhabi, United Arab Emirates, pp. 983-987, 2020. (Oral)
8. **Zhiyuan Zha**, Bihan Wen, Jiachao Zhang, Jiantao Zhou and Ce Zhu. "A Comparative Study for the Nuclear Norms Minimization Methods", in *26th IEEE International Conference on Image Processing (ICIP)*, Taipei, Taiwan, China, pp. 2050-2054, 2019. (Poster)
9. **Zhiyuan Zha**, Xin Yuan, Bihan Wen, Jiachao Zhang, Jiantao Zhou and Ce Zhu. "Simultaneous Nonlocal Self-Similarity Prior for Image Denoising", in *26th IEEE International Conference on Image Processing (ICIP)*, Taipei, Taiwan, China, pp. 1119-1123, 2019. (Poster)
10. **Zhiyuan Zha**, Xinggan Zhang, Qiong Wang, Yechao Bai, Lan Tang and Xin Yuan. "Group sparsity residual with non-local samples for image denoising", in *43th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Calgary, AB, Canada, pp. 1353-1357, 2018. (Oral)
11. **Zhiyuan Zha**, Xin Yuan, Bihan Wen, Jiantao Zhou and Ce Zhu. "Joint patch-group based sparse representation for image inpainting", in *10th Asian Conference on Machine Learning (ACML)*, Beijing, China, pp. 145-160. 2018. (Oral)
12. **Zhiyuan Zha**, Xin Liu, Xiaohua Huang, Henglin Shi, Yingyue Xu, Qiong Wang, Lan Tang and Xinggan Zhang, "Analyzing the Group Sparsity based on the Rank Minimization Methods". in *17th IEEE International Conference on Multimedia and Expo (ICME)*, Hong Kong, China, pp. 883-888, 2017. (**Platinum Best Paper Award, Top 0.5%, Total 850**, Oral)
13. **Zhiyuan Zha**, Xin Liu, Ziheng Zhou, Xiaohua Huang, Jingang Shi, Zhenhong Shang, Lan Tang, Yechao Bai, Qiong Wang and Xinggan Zhang. "Image denoising via group sparsity residual constraint", in *42th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, New Orleans, LA, USA, pp. 1787-1791, 2017. (Poster)
14. **Zhiyuan Zha**, Xinggan Zhang, Qiong Wang, Yechao Bai and Lan Tang. "Image denoising using group sparsity residual and external nonlocal self-similarity prior", in *24th IEEE International Conference on Image Processing (ICIP)*, Beijing, China, pp. 2956-2960. IEEE, 2017. (Poster)
15. Lanqing Guo, **Zhiyuan Zha**, Saiprasad Ravishankar, and Bihan Wen. "Self-convolution: A highly-efficient operator for non-local image restoration", in *46th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Toronto, Ontario, Canada, pp. 1860-1864, 2021. (Oral)

PROGRAM COMMITTEE OR REVIEWER

- International Conference on Learning Representations (ICLR)
- AAI Conference on Artificial Intelligence (AAAI)
- ACM Multimedia (ACM MM)
- IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)
- IEEE International Conference on Image Processing (ICIP)
- IEEE International Conference on Multimedia and Expo (ICME)
- IEEE International Symposium on Biomedical Imaging (ISBI)
- IEEE International Symposium on Circuits & Systems (ISCAS)

JOURNAL REVIEWER

- IEEE Signal Processing Magazine
- IEEE Transactions on Pattern Analysis and Machine Intelligence
- International Journal of Computer Vision
- IEEE Transactions on Image Processing
- IEEE Transactions on Signal Processing
- IEEE Transactions on Neural Networks and Learning Systems

- IEEE Transactions on Multimedia
- IEEE Transactions on Circuits and Systems for Video Technology
- IEEE Transactions on Medical Imaging
- IEEE Transactions on Geoscience and Remote Sensing
- IEEE Transactions on Computational Imaging
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Broadcasting
- IEEE Signal Processing Letters
- ACM Transactions on Multimedia Computing, Communications, and Applications
- SIAM Journal on Imaging Sciences
- Inverse Problems
- Pattern Recognition
- Information Fusion
- Neurocomputing
- Signal Processing
- Signal Processing: Image Communication
- Digital Signal Processing
- Journal of Visual Communication and Image Representation
- The Visual Computer
- IET Image Processing
- CAAI Transactions on Intelligence Technology